BRAINWARE UNIVERSITY

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

Assignment Submission

Assignment No: - 02

Name of the Department:- Computer Science and Engineering

Programme Name: - BTech in CSE

Semester / year:- 3rd semester / 2nd year

Course Code: - BTS30102

Course Name: - Database Management Systems

Answer the following Questions:-

<table>
<tr><td></td><td></td><td></td><td>CO</td></tr>
<tr><td>Question no 1</td><td>:</td><td>Case Study: A bank maintains customer accounts through a computerized database system. One account, (A), has an initial balance of ₹10,000. Two transactions are executed concurrently: transaction (T_1) withdraws ₹2,000, while transaction (T_2) deposits ₹3,000. Due to concurrent execution, both transactions read and update the same account balance. The schedule is:

<table>
<tr><th>Step</th><th>Transaction T_1</th><th>Transaction T_2</th></tr>
<tr><td>1</td><td>$R_1(A)$</td><td></td></tr>
<tr><td>2</td><td></td><td>$R_2(A)$</td></tr>
<tr><td>3</td><td>$A = A - 2000$</td><td></td></tr>
<tr><td>4</td><td></td><td>$A = A + 3000$</td></tr>
<tr><td>5</td><td>$W_1(A)$</td><td></td></tr>
<tr><td>6</td><td></td><td>$W_2(A)$</td></tr>
</table>
(a). Analyze the schedule and identify the conflicting operations. Construct its precedence graph and determine whether the schedule is conflict-serializable.

(b) Evaluate the final balance produced by the given schedule. Compare it with the correct balance obtained when both transactions execute serially, and analyze the inconsistency.</td><td>CO4
CO5</td></tr>
<tr><td>Question no 2</td><td>:</td><td>Case Study: A bank maintains 10,000 customer account records in its database. Each disk block can store 100 account records. To improve the efficiency of account searches, the bank implements a B-tree index in which each internal node can have a maximum of five children. The database administrator wants to compare indexed searching with</td><td>CO4
CO5</td></tr>
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		sequential scanning to determine the most efficient approach for retrieving an individual account. Answer the following questions: (a) Analyze the number of disk blocks required to store all account records and determine the minimum number of leaf-level index entries required if each record has one index entry. Also, calculate the maximum number of children in a B-tree of height 3. (b) Two indexing strategies are considered: a B-tree index with a maximum of 5 children per internal node and a sequential scan of all 10,000 records. Evaluate which strategy is more efficient for searching a single account, assuming one disk access per tree level and 100 records per disk block for the sequential scan.	
Question no 3		**Case Study:** A hospital uses a database to maintain patient records. Three transactions attempt to access the same patient record (P). Transaction (T_1) obtains an exclusive lock on (P) at time 10 and releases it at time 30. Transaction (T_2) requests a shared lock on (P) at time 15 to view the patient's medical history, while transaction (T_3) requests an exclusive lock at time 20 to update the patient's treatment details. The database grants locks only when they are compatible with existing locks. Answer the following questions: (a) Analyze the lock compatibility and determine the earliest possible time at which (T_2) and (T_3) can obtain their requested locks, assuming (T_1) releases its lock at time 30 and each granted lock is released immediately after use. (b) Evaluate whether strict two-phase locking is suitable for this hospital database. Compare its benefits in maintaining recoverability with the possible waiting time and concurrency limitations illustrated in the case.	CO4 CO5
Question no 4		**Case Study:** An online shopping platform maintains product inventory and sales information in a database. Two concurrent transactions, (T_1) and (T_2), access and update the inventory items (X) and (Y). Transaction (T_1) processes a customer's order, while transaction (T_2) updates the inventory for another order. Their concurrent schedule is:	CO4 CO5

Step	Transaction T_1	Transaction T_2
1	$R_1(X)$	
2		$R_2(X)$
3	$W_1(X)$	
4		$W_2(X)$
5	$R_1(Y)$	
6		$R_2(Y)$
7	$W_1(Y)$	
8		$W_2(Y)$

(a) Construct the precedence graph for the given schedule. Identify all conflicting operations and determine whether the schedule is conflict-serializable.

(b) Evaluate the given schedule for correctness and determine whether it is equivalent to either serial execution order (T_1 -> T_2) or (T_2 -> T_1). Justify your conclusion using the precedence graph.

Question no 5		**Case Study:** A college operates an online portal through which students and faculty members access academic records. The portal has 2,000 registered users and receives 500 login attempts on a particular day. Of these, 420 attempts are successful and 80 fail. To prevent unauthorized access, the system locks an account after five consecutive failed login attempts. The security administrator wants to assess the effectiveness of this policy and determine whether additional authentication mechanisms are required. Answer the following questions: (a) Calculate the login success rate and failure rate as percentages. If 30 failed attempts belong to one account and occur in six groups of five consecutive failures, analyze how many account-lockout events may occur under the stated policy, assuming each group triggers a separate lockout. (b) Evaluate the effectiveness of the five-failure lockout policy. Recommend whether account lockout alone is sufficient or should be supplemented with multifactor authentication and rate limiting. Justify your recommendation using the case statistics.	CO4 CO5

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Name of the Subject Teacher: Dr. Suman Das
Designation: Associate Professor
Date of Notification: 29.10.2026
Date of Examination: 05.11.2026